

APPLIED MATH 3

WEEK 6

EASING INTO BETTER GAMEPLAY

An introduction to linear equations and Lerp

SLOPE INTERCEPT FORM

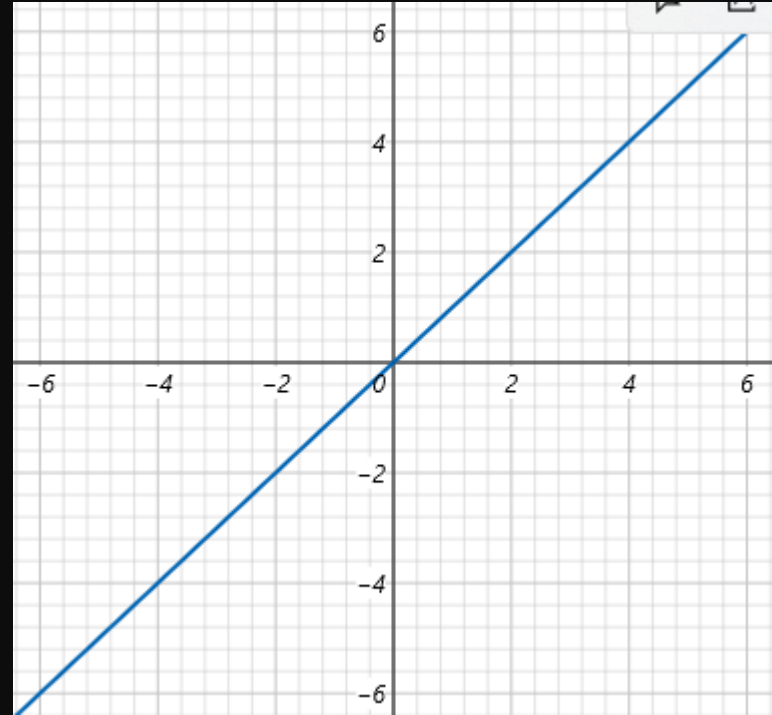
- When describing a line mathematically, there are multiple formulas that can be used. One particularly useful one is the Slope-Intercept form.
- This is the graph of a one variable linear equation, in this case X is the variable.
- Any equation of this general form will give us a straight line
 - (Hence: Linear Equations)

$$y = mx + b$$

- Where Y is the Y axis value
- X is the X axis value
- M is the slope
- B is the Y-Intercept of the line (where the line crosses the Y Axis).

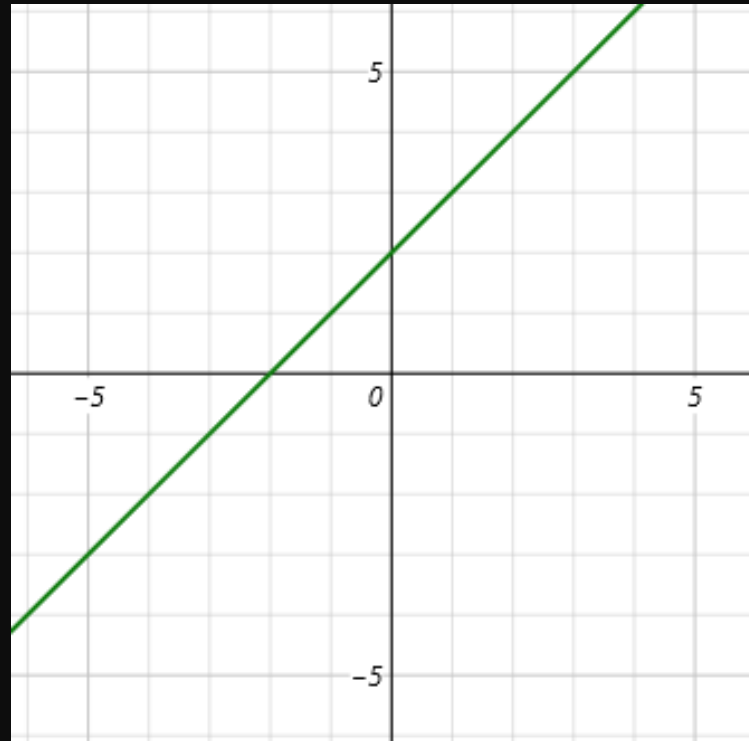
SLOPE INTERCEPT FORM

- Basic Graph Example
 $y = (1)x + 0$



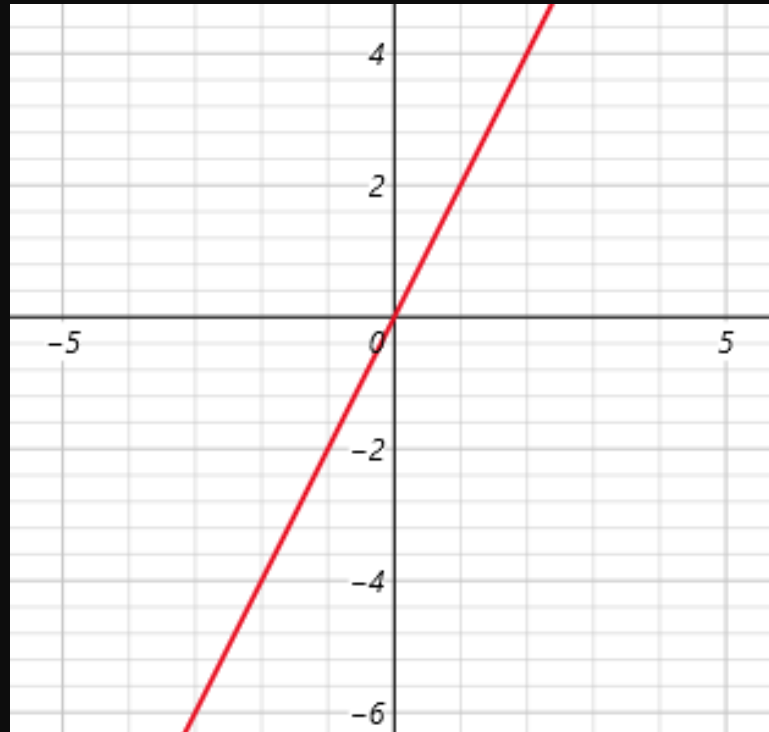
SLOPE INTERCEPT FORM

- Y intercept
 $y = (1)x + 2$



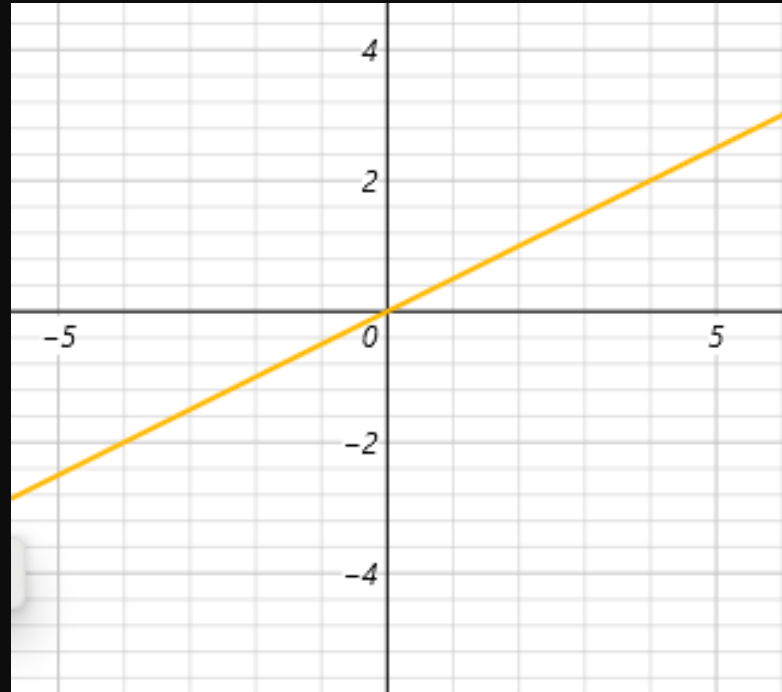
SLOPE INTERCEPT FORM

- Adjust Slope
 $y = (2)x + 0$



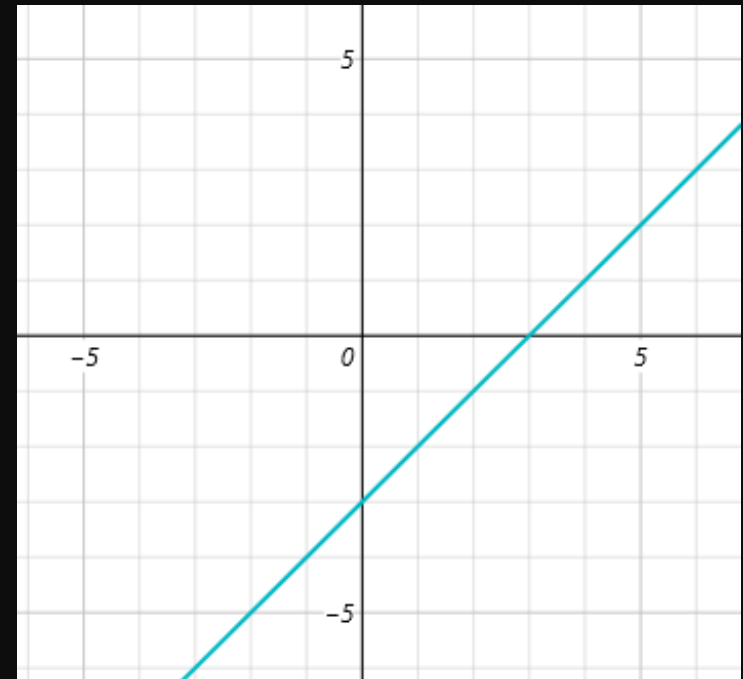
SLOPE INTERCEPT FORM

- Adjust Slope again
 $y = (1/2)x + 0$



SLOPE INTERCEPT FORM

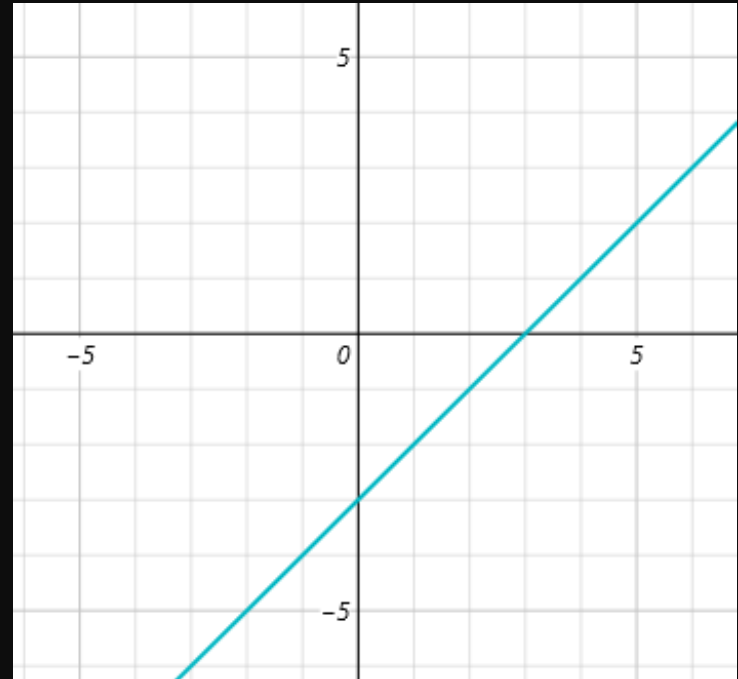
- Y-Intercept can be negative as well
 $y = (1)x - 3$



SLOPE INTERCEPT FORM

- Any real number can be used,
- Combining both slope and Y-intercept adjustment

$$y = (1.4)x + 3$$



SLOPE INTERCEPT FORM

- Exercise: Use the Graphing calculator in window to create some of your own equations, adjusting the M and B values

SLOPE ADJUSTMENTS: DESIGNING FUNCTIONS

- Using these basic tools, we can begin to think about designing functions for use in our games.
- These could be anything from response curves, to scaling stats based on current level. Understanding how to create equations such as these is a very useful tool.
- For the more visually minded, this is exactly what we are doing whenever we are creating curves for transitions, animation timelines and particle system properties, however the game engines are providing use with a handy visual interface.

ASIDE: USEFULNESS OF NORMALIZATION

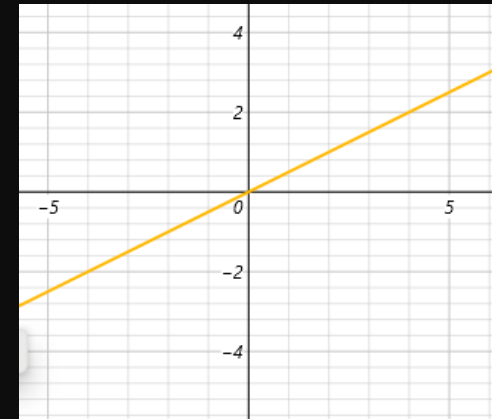
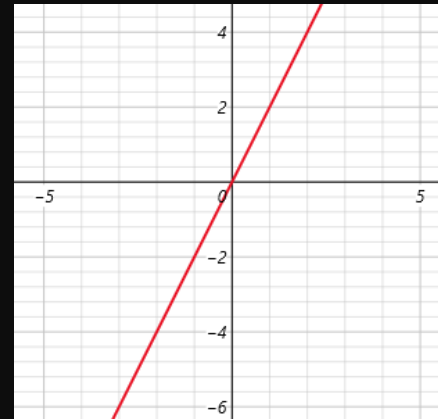
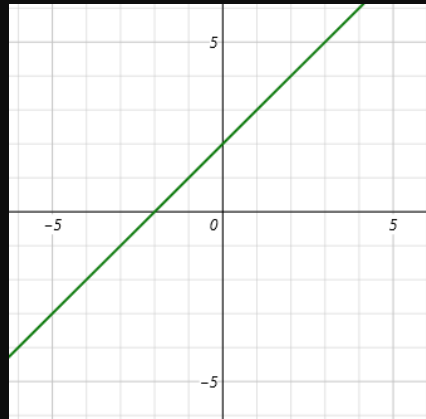
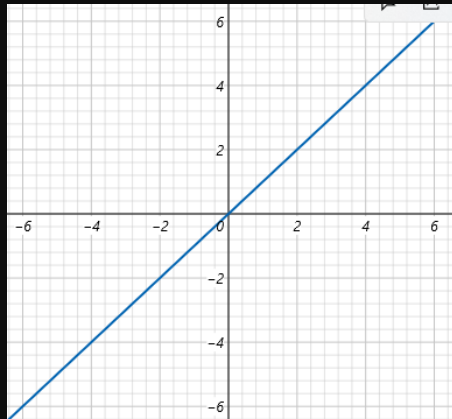
- 0 to 1

ASIDE: MAPPING DATA

- New Ranges, same distribution

EXAMPLE EASING: LINEAR

- Straight Path
- These are the Linear equations we saw earlier.
- Typically Time is the X variable, and whatever property is being eased is the Y

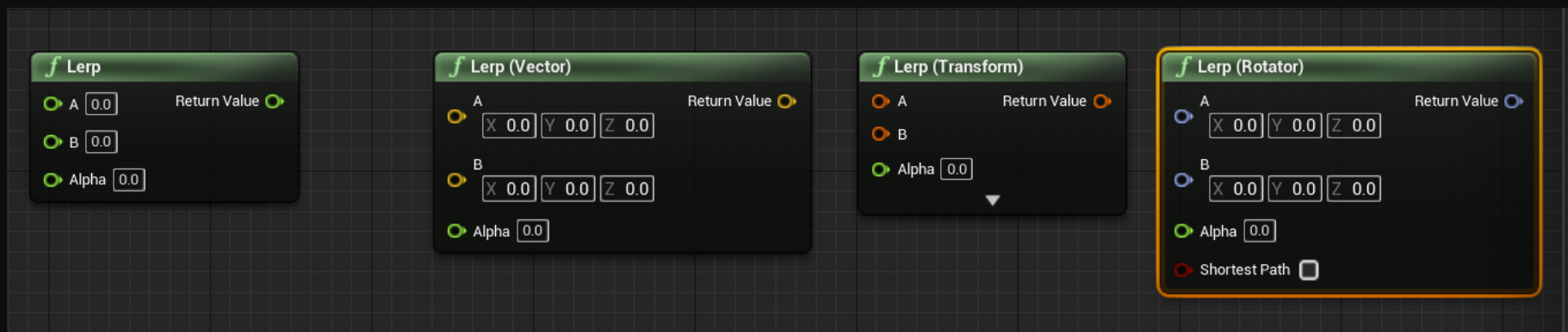


INTERPOLATION NODES IN UNREAL

- When making our games, we often need to move or rotate an object from one position/rotation to another.
- We have several options for how to do this and each gives us a different gameplay effect and “feel”
- Understanding these concepts is the key to having your movement smooth and controlled. It also allows you to adjust the “Feeling” of movement.
- We will examine movement first, as it is simpler to understand, however the same principles apply to rotations.

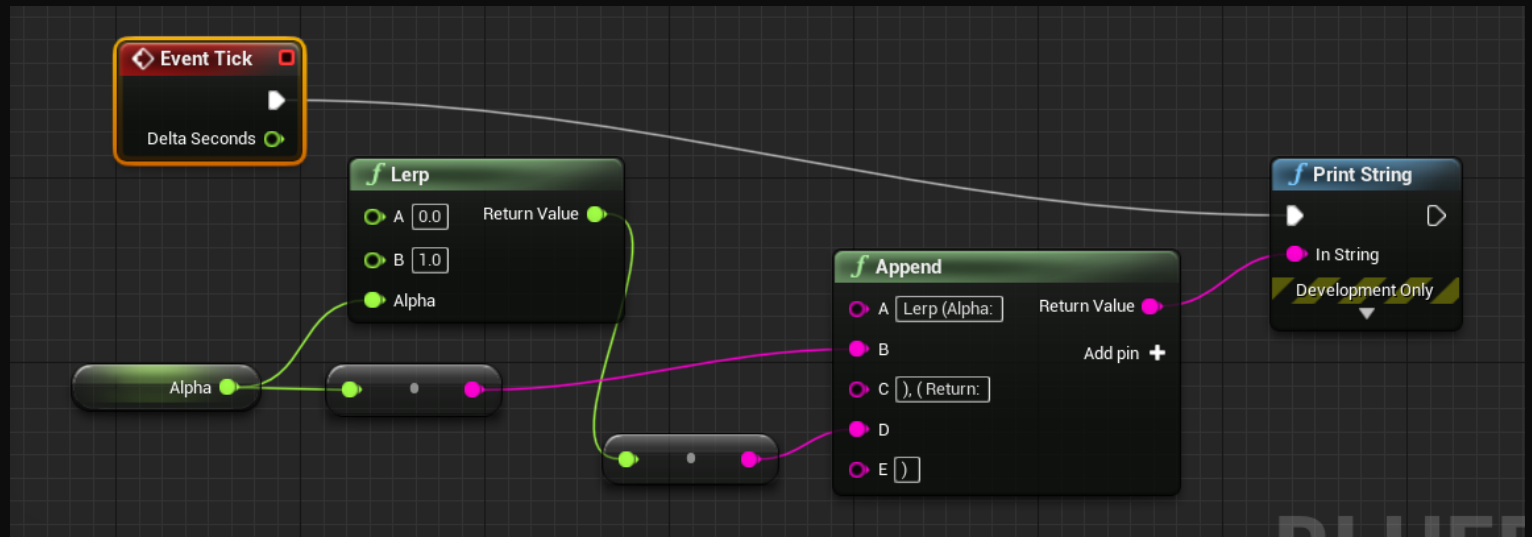
USEFUL UNREAL NODES: LERP

- The mysterious Lerp node can be tricky to get your head around at first glance, however it is very powerful and useful tool for controlling any number of different movements, rotations and transitions within unreal.
- A Lerp is short for “Linear Interpolation” and as we have seen, it is useful for transitioning between two states.
- There are several varieties of Lerp nodes, each dealing with a different data type, however they all work under the same principles.



USEFUL UNREAL NODES: LERP

- Basically, the lerp node accepts a starting value and an ending value (A & B) as well as the Alpha.
- The alpha specifies what proportion of the transition to display.
- Using a float for example, our simple example demonstrates what is really happening.

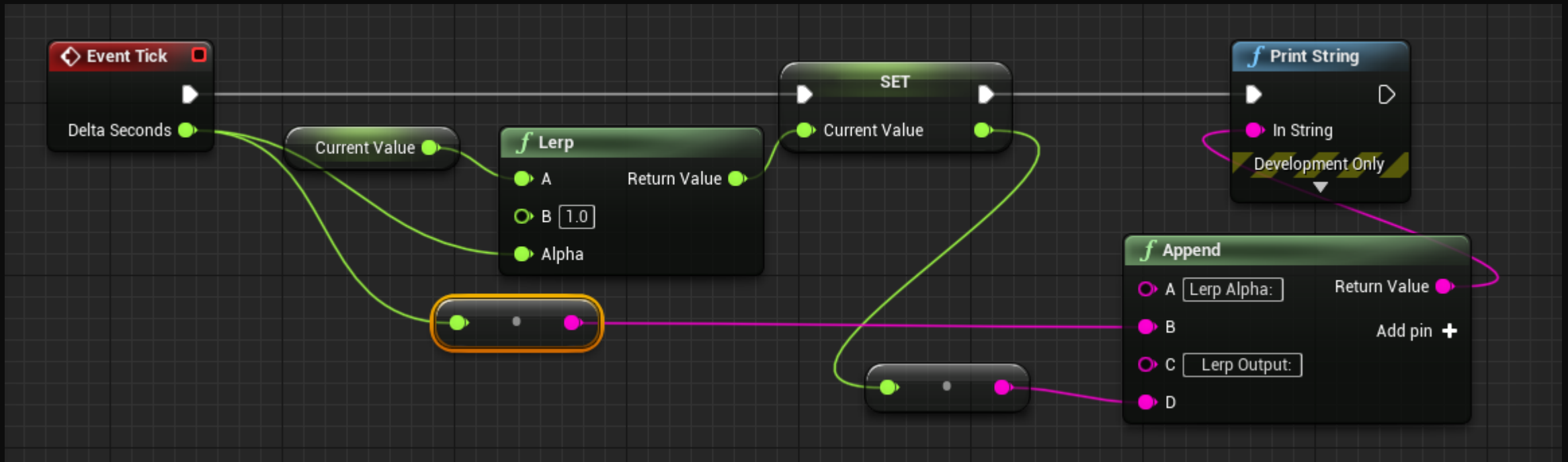


USEFUL UNREAL NODES: LERP

- Experiment with this example and what you will find is:
 - When the value of Alpha is 0, the output is 0
 - When the value of Alpha is 1, the output is 1
 - When the value of Alpha is 0.5, the output is 0.5
- Simply put, the Lerp is finding an interpolation of the range (A,B)
- An alpha of .5 means give us the value 50% between A and B
- An alpha of .75 means give us the value 75% between A and B, or .75 in this example.

USEFUL UNREAL NODES: LERP

- You might find this confusing as you often see a Lerp used with an input value of DeltaTime, something like this: To understand what this is really doing however, just think about what we talked about earlier.



USEFUL UNREAL NODES: LERP

- Here is some partial output for reference..
- The behaviour of this is that it will start updating the value quite quickly, and gradually slow to smaller and smaller output steps, approaching our B target vector but not *Really* reaching it.

[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.016667
[LerpBasics_2]	Lerp Alpha: 0.116648	Lerp Output: 0.131371
[LerpBasics_2]	Lerp Alpha: 0.063707	Lerp Output: 0.186709
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.200264
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.213593
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.2267
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.239589
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.252262
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.264725
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.27698
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.28903
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.30088
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.312532
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.32399
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.335257
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.346336
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.357231
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.367944
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.378479
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.388837
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.399024
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.40904
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.41889
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.428575
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.438099
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.447464
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.456673
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.465729
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.474633
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.48339
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.492
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.500467
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.508793
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.516979
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.52503
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.532946
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.54073
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.548385
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.555912
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.563314
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.570592

USEFUL UNREAL NODES: LERP

- This is more of the log as it approaches 1
- Note it doesn't actually reach the value of 1
 - (It will eventually but it will take a while!)
- This is similar to (although not exactly) the Logarithmic curve we examined earlier.

[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.978832
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.979185
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.979532
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.979873
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.980208
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.980538
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.980863
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.981182
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.981495
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.981804
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.982107
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.982405
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.982698
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.982987
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.98327
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.983549
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.983823
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.984093
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.984358
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.984619
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.984875
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.985127
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.985375
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.985619
[LerpBasics_2]	Lerp Alpha: 0.01667	Lerp Output: 0.985859
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.986094
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.986326
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.986554
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.986778
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.986998
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.987215
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.987428
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.987638
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.987844
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.988046
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.988246
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.988441
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.988634
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.988824
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.98901
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.989193
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.989373
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.98955
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.989724
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.989896
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.990064

USEFUL UNREAL NODES: LERP

- To Demonstrate what is happening here, I will use a simplified example with smaller time steps. Imagine a Lerp node with a value of $A = 0$, $B=1$. That is we are Lerping from 0 to 1. (We could use any values here but this is much simpler to understand). Imagine our Lerp Alpha (timestep) is 0.1
- At timestep 0 (t_0) our output will be 0, since the lerp has not been applied yet.
- At T_1 , our output will be 0.1, since we are .1 of the way from A to B.
- Since we are saving our output value back into our variable (that feeds into A), when we take T_2 , our A value will be 0.1
- This means that even though the Alpha stays the same, the range it is lerping over has changes from 0..1 to (0.1... 1).
- 0.1 of that range is 0.19, or 10% of the way from $A = 0.1$ to $B = 1.0$

USEFUL UNREAL NODES: LERP

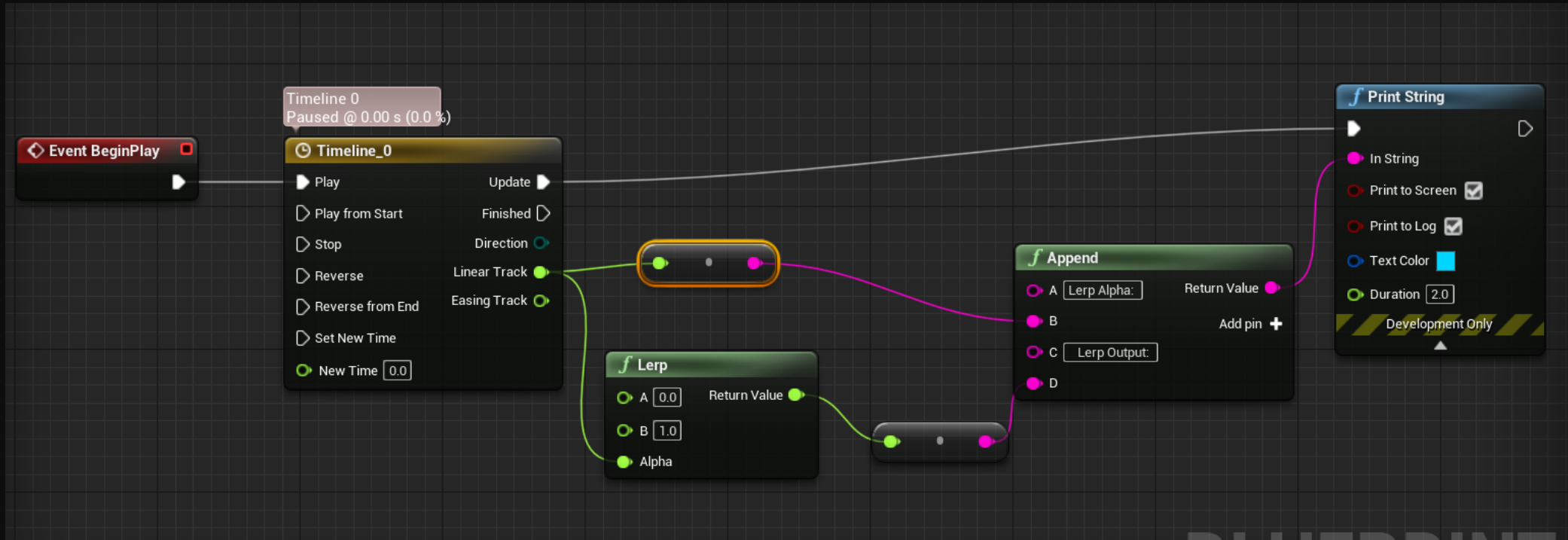
- If we continue running our experiment, we can see that we have a similar effect to what we had before.
- Question: if you don't save the output, what would you get?
 - The output would be 0.1 every frame since A would remain 0

Lerp Alpha: 0.1	Lerp Output: 0.1
Lerp Alpha: 0.1	Lerp Output: 0.19
Lerp Alpha: 0.1	Lerp Output: 0.271
Lerp Alpha: 0.1	Lerp Output: 0.3439
Lerp Alpha: 0.1	Lerp Output: 0.40951
Lerp Alpha: 0.1	Lerp Output: 0.468559
Lerp Alpha: 0.1	Lerp Output: 0.521703
Lerp Alpha: 0.1	Lerp Output: 0.569533
Lerp Alpha: 0.1	Lerp Output: 0.61258
Lerp Alpha: 0.1	Lerp Output: 0.651322
Lerp Alpha: 0.1	Lerp Output: 0.686189
Lerp Alpha: 0.1	Lerp Output: 0.71757
Lerp Alpha: 0.1	Lerp Output: 0.745813
Lerp Alpha: 0.1	Lerp Output: 0.771232
Lerp Alpha: 0.1	Lerp Output: 0.794109
Lerp Alpha: 0.1	Lerp Output: 0.814698
Lerp Alpha: 0.1	Lerp Output: 0.833228
Lerp Alpha: 0.1	Lerp Output: 0.849905
Lerp Alpha: 0.1	Lerp Output: 0.864915
Lerp Alpha: 0.1	Lerp Output: 0.878423
Lerp Alpha: 0.1	Lerp Output: 0.890581
Lerp Alpha: 0.1	Lerp Output: 0.901523
Lerp Alpha: 0.1	Lerp Output: 0.911371
Lerp Alpha: 0.1	Lerp Output: 0.920234
Lerp Alpha: 0.1	Lerp Output: 0.92821
Lerp Alpha: 0.1	Lerp Output: 0.935389
Lerp Alpha: 0.1	Lerp Output: 0.94185
Lerp Alpha: 0.1	Lerp Output: 0.947665
Lerp Alpha: 0.1	Lerp Output: 0.952899
Lerp Alpha: 0.1	Lerp Output: 0.957609
Lerp Alpha: 0.1	Lerp Output: 0.961848

USEFUL UNREAL NODES: LERP

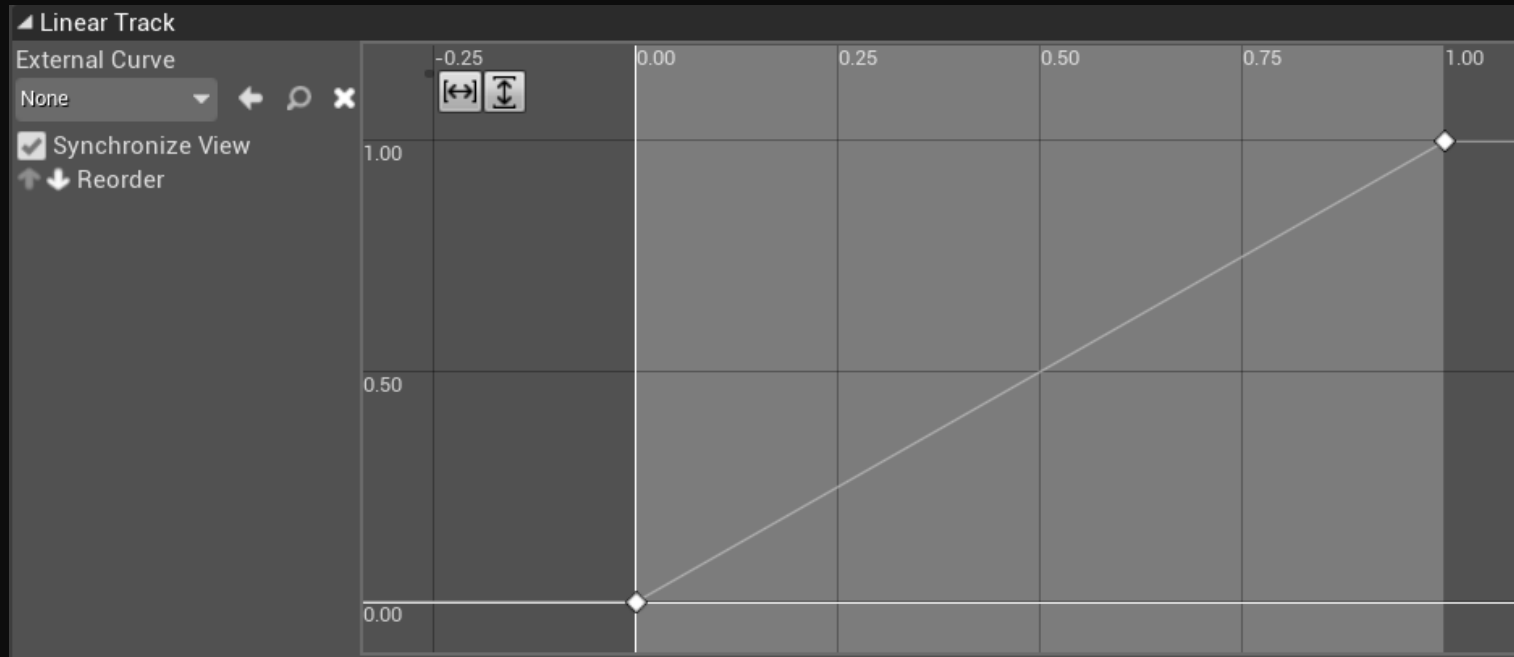
- In summary, Lerp often gets abused and misused. It is not a great option for using with deltaTime since it never quite reaches its target and there are better options for easing (as we will see).
- What lerp is extremely useful for is manually interpolating between two values, finding a midpoint or proportion of two values.
- While this can be used in many different ways, one of the most powerful ways of using this in Unreal is to combine the Lerp node with a Timeline. This way, you can control the easing precisely.
 - (Demo on next slide).

USEFUL UNREAL NODES: LERP WITH TIMELINE



- This might not make much sense. Why use a lerp at all if you already have a timeline?
- You are correct in this example because our Lerp is always from 0..1, however, normally it is between two different, or even changing values so the Lerp node is simply allowing the timeline to work on any start and end value.

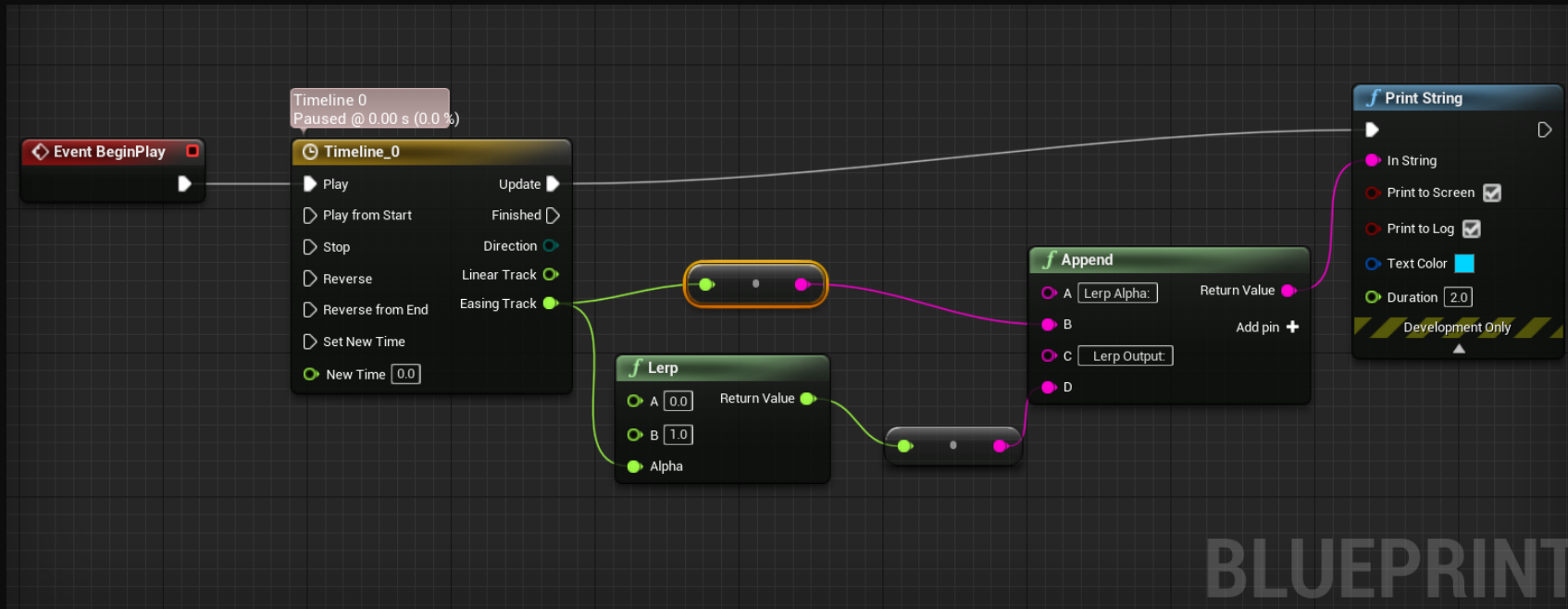
USEFUL UNREAL NODES: LERP WITH TIMELINE



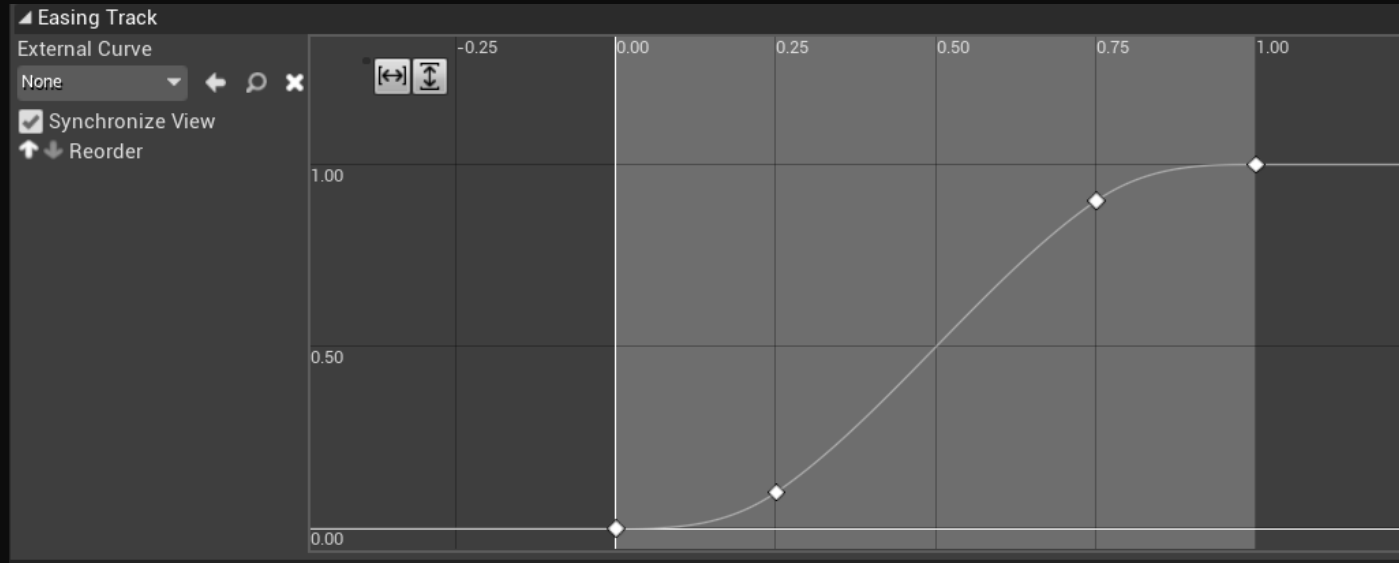
Lerp Alpha: 0.016667	Lerp Output: 0.016667
Lerp Alpha: 0.134483	Lerp Output: 0.134483
Lerp Alpha: 0.151115	Lerp Output: 0.151115
Lerp Alpha: 0.167817	Lerp Output: 0.167817
Lerp Alpha: 0.184483	Lerp Output: 0.184483
Lerp Alpha: 0.201115	Lerp Output: 0.201115
Lerp Alpha: 0.217817	Lerp Output: 0.217817
Lerp Alpha: 0.234484	Lerp Output: 0.234484
Lerp Alpha: 0.251151	Lerp Output: 0.251151
Lerp Alpha: 0.267818	Lerp Output: 0.267818
Lerp Alpha: 0.284485	Lerp Output: 0.284485
Lerp Alpha: 0.301152	Lerp Output: 0.301152
Lerp Alpha: 0.317819	Lerp Output: 0.317819
Lerp Alpha: 0.334485	Lerp Output: 0.334485
Lerp Alpha: 0.351152	Lerp Output: 0.351152
Lerp Alpha: 0.367819	Lerp Output: 0.367819
Lerp Alpha: 0.384486	Lerp Output: 0.384486
Lerp Alpha: 0.401153	Lerp Output: 0.401153
Lerp Alpha: 0.41782	Lerp Output: 0.41782
Lerp Alpha: 0.434487	Lerp Output: 0.434487
Lerp Alpha: 0.451154	Lerp Output: 0.451154
Lerp Alpha: 0.467821	Lerp Output: 0.467821
Lerp Alpha: 0.484488	Lerp Output: 0.484488
Lerp Alpha: 0.501155	Lerp Output: 0.501155
Lerp Alpha: 0.517822	Lerp Output: 0.517822
Lerp Alpha: 0.534488	Lerp Output: 0.534488
Lerp Alpha: 0.551155	Lerp Output: 0.551155
Lerp Alpha: 0.567822	Lerp Output: 0.567822
Lerp Alpha: 0.584489	Lerp Output: 0.584489

USEFUL UNREAL NODES: LERP WITH EASING TIMELINE

AS YOU CAN SEE, THIS GIVES US THE EASING WE WANT WHILE MAINTAINING SPECIFIC CONTROL OVER HOW IT FUNCTIONS. THIS EASING TRACK HAS A BIT OF AN S-CURVE SHAPE THAT WILL GIVE US AND EASE-IN AND EASE-OUT TYPE BEHAVIOUR.



USEFUL UNREAL NODES: LERP WITH EASING TIMELINE



Lerp Alpha: 0.00003	Lerp Output: 0.00003
Lerp Alpha: 0.085473	Lerp Output: 0.085473
Lerp Alpha: 0.122855	Lerp Output: 0.122855
Lerp Alpha: 0.144876	Lerp Output: 0.144876
Lerp Alpha: 0.168042	Lerp Output: 0.168042
Lerp Alpha: 0.192266	Lerp Output: 0.192266
Lerp Alpha: 0.217459	Lerp Output: 0.217459
Lerp Alpha: 0.243531	Lerp Output: 0.243531
Lerp Alpha: 0.270394	Lerp Output: 0.270394
Lerp Alpha: 0.29796	Lerp Output: 0.29796
Lerp Alpha: 0.326138	Lerp Output: 0.326138
Lerp Alpha: 0.35484	Lerp Output: 0.35484
Lerp Alpha: 0.384021	Lerp Output: 0.384021
Lerp Alpha: 0.413506	Lerp Output: 0.413506
Lerp Alpha: 0.443248	Lerp Output: 0.443248
Lerp Alpha: 0.473169	Lerp Output: 0.473169
Lerp Alpha: 0.503158	Lerp Output: 0.503158
Lerp Alpha: 0.533139	Lerp Output: 0.533139
Lerp Alpha: 0.563022	Lerp Output: 0.563022
Lerp Alpha: 0.592717	Lerp Output: 0.592717
Lerp Alpha: 0.622136	Lerp Output: 0.622136
Lerp Alpha: 0.651191	Lerp Output: 0.651191
Lerp Alpha: 0.679791	Lerp Output: 0.679791
Lerp Alpha: 0.707849	Lerp Output: 0.707849
Lerp Alpha: 0.735275	Lerp Output: 0.735275
Lerp Alpha: 0.761981	Lerp Output: 0.761981
Lerp Alpha: 0.787877	Lerp Output: 0.787877
Lerp Alpha: 0.812875	Lerp Output: 0.812875
Lerp Alpha: 0.836885	Lerp Output: 0.836885
Lerp Alpha: 0.85982	Lerp Output: 0.85982
Lerp Alpha: 0.881589	Lerp Output: 0.881589
Lerp Alpha: 0.902098	Lerp Output: 0.902098
Lerp Alpha: 0.935442	Lerp Output: 0.935442
Lerp Alpha: 0.949259	Lerp Output: 0.949259
Lerp Alpha: 0.960949	Lerp Output: 0.960949
Lerp Alpha: 0.970689	Lerp Output: 0.970689
Lerp Alpha: 0.978658	Lerp Output: 0.978658
Lerp Alpha: 0.985034	Lerp Output: 0.985034
Lerp Alpha: 0.989993	Lerp Output: 0.989993
Lerp Alpha: 0.993715	Lerp Output: 0.993715
Lerp Alpha: 0.996376	Lerp Output: 0.996376
Lerp Alpha: 0.998155	Lerp Output: 0.998155
Lerp Alpha: 0.999229	Lerp Output: 0.999229
Lerp Alpha: 0.999776	Lerp Output: 0.999776
Lerp Alpha: 0.999973	Lerp Output: 0.999973
Lerp Alpha: 1.0	Lerp Output: 1.0

USEFUL UNREAL NODES: LERP

- As you can see, Lerp is extremely useful and can be used to interpolate between any number of different things, including Vectors, rotations and entire transforms.
- It is also one of the most likely to be misunderstood and misused so it is important to have a good idea of what it is doing
- It's also quite simple to understand once you get the basic concept
- Next we will look at some alternative nodes in Unreal for easing between different states.

USEFUL UNREAL NODES: LERP

- While Lerp will move you gradually closer to a point, there are some issues using it for this effect.
 - 1. It can be hard to control the rate at which it reaches its goal
 - 2. It doesn't quite reach the goal value, when tends to leave it sort of jittering as it inches towards it.
- As we will soon see, there are alternative nodes that achieve a similar effect but are better suited to this purpose.